Red Blood Cells

Author: Usense Healthcare

Identity Sheet

Property	Value
Name	Red Blood Cells (Erythrocytes)
Other names	RBCs, erythrocytes, red cells
Nature	Anucleate biconcave disc cells
Diameter	6–8 µm
Thickness	~2 µm (edge), ~1 µm (centre)
Mean cell volume (MCV)	80–100 fL
Main component	Haemoglobin (~270 million molecules/cell, ~33% of cell mass)
Lifespan	~120 days in circulation

Morphology:

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Side view:  )( Biconcave disc
Top view:  O  7 um diameter
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Red blood cells are the most abundant cells in blood (~4.5–5.5 million/µL). They carry haemoglobin for oxygen transport. In urine, their presence (haematuria) is always abnormal above trace quantities and indicates bleeding somewhere in the urinary tract. RBCs in urine may be **isomorphic** (uniform, suggesting lower urinary tract origin) or **dysmorphic** (distorted, acanthocytes, suggesting glomerular origin).

Molecules Not to Be Confused With

Entity	Description	Key Difference
Free haemoglobin	Hb released from lysed RBCs	Dissolved protein, no cells visible on microscopy; dipstick positive
Myoglobin	Muscle protein (MW 17,800 Da)	Similar dipstick reaction; distinguished by serum colour and CK levels
White blood cells	Leukocytes, 10–15 µm, granular	Larger, nucleated, different microscopic appearance
Yeast cells	*Candida*, 3–8 µm, budding	Oval, budding, no central pallor

Medical Information

Origin

Endogenous: Erythropoiesis RBCs are produced in the **bone marrow** (adults) from haematopoietic stem cells through erythropoiesis, stimulated by **erythropoietin (EPO)** from the kidneys. Stages: proerythroblast, basophilic,

polychromatic, orthochromatic erythroblast, reticulocyte (enucleation), mature RBC. Production rate: ~200 billion RBCs/day. The process requires iron, vitamin B12, folate, and EPO.

Exogenous RBCs are not obtained exogenously under normal conditions. Blood transfusion introduces donor RBCs. No dietary source of intact RBCs exists.

Primary & Secondary Biological Roles

Primary role: - **Oxygen transport:** Haemoglobin in RBCs binds O_2 in lungs and delivers it to tissues. - **CO_2 transport:** RBCs carry ~20% of CO_2 as carbaminohaemoglobin; carbonic anhydrase converts CO_2 to HCO_3^-.

Secondary roles: - **pH buffering:** Haemoglobin acts as a buffer via histidine residues. - **Nitric oxide metabolism:** Deoxyhaemoglobin reduces nitrite to NO, contributing to vasodilation. - **Blood viscosity:** RBCs are the primary determinant of blood viscosity.

Catabolism and Elimination Pathway

- Aged/damaged RBCs are removed by **reticuloendothelial macrophages** in the spleen, liver, and bone marrow.
- Haemoglobin is broken down: globin → amino acids; heme → biliverdin → bilirubin (excreted in bile); iron → recycled via transferrin.
- RBCs should **not be eliminated in urine** — their presence indicates pathology.
- Normal glomerular filtration barrier (endothelium + basement membrane + podocytes) prevents RBC passage.

Expression in Humans

Normal Levels

Compartment	Reference Range
Blood RBC count	Male: 4.5–5.5 × 10^12/L; Female: 4.0–5.0 × 10^12/L
Urine RBCs (microscopy)	0–2 RBC/HPF (high-power field); <5 RBC/HPF considered normal
Urine RBCs (flow cytometry)	<25 RBC/µL (Sysmex UF)
Dipstick blood	Negative (trace may be acceptable)

Factors Influencing Levels **Increased RBCs in urine (haematuria):** - Urinary tract infection - Nephrolithiasis (kidney stones) - Glomerulonephritis (IgA nephropathy, post-streptococcal) - Bladder/renal/prostate cancer - Trauma, vigorous exercise - Menstrual contamination - Anticoagulant therapy - Polycystic kidney disease

Decreased / absent (normal state): - Healthy urinary tract - Adequate hydration - No structural or inflammatory urinary tract pathology

Associated Pathologies

Condition	RBC Pattern	Key Symptoms
IgA nephropathy	Dysmorphic RBCs, RBC casts	Episodic gross haematuria with URI; most common glomerulonephritis worldwide
Nephrolithiasis	Isomorphic RBCs	Flank pain, colicky, +/- gross haematuria; prevalence ~10% lifetime

Condition	RBC Pattern	Key Symptoms
Bladder cancer	Isomorphic, painless gross haematuria	Risk factors: smoking, age >50; ~550,000 new cases/year globally
UTI	Isomorphic + bacteria + WBCs	Dysuria, frequency
Exercise-induced haematuria	Transient, resolves 24–72 h	Benign; occurs in ~20% of runners
Glomerulonephritis	Dysmorphic RBCs, acanthocytes >5%, RBC casts	Proteinuria, oedema, hypertension

Presence in Urine

Should it be normally present? No — only trace amounts (0–2/HPF). More than 3–5 RBC/HPF constitutes microscopic haematuria.

Normal urinary levels: <5 RBC/HPF on microscopy; dipstick negative.

Form in urine: Intact cells — erythrocytes in their native biconcave form (isomorphic) or distorted form (dysmorphic/acanthocytes from glomerular passage). In hypotonic or alkaline urine, RBCs may lyse → "ghost cells" releasing free haemoglobin.

Pathological significance:

Urinary RBCs	Possible Causes	Prevalence
Microscopic haematuria (>3 RBC/HPF)	Stones, infection, cancer, glomerular disease, exercise	~2–31% of general population
Gross haematuria (visible)	Cancer, stones, severe UTI, glomerulonephritis, trauma	~0.5% of adults presenting
Absent	Normal	—

Solubility: Not applicable — intact cells, not dissolved.

Detection in Urine

Available Clinical Assays

1. **Urine dipstick (chemical detection of haemoglobin):**
   - **Principle:** Pseudoperoxidase activity of haemoglobin catalyses oxidation of tetramethylbenzidine (TMB) by organic peroxide → green-blue colour.
   - **Detection:** Colour change on reagent pad graded 0 to 3+ (or trace to large). Detects both intact RBCs (spotted pattern) and free Hb (uniform colour).
   - **LOD:** ~5–10 RBC/µL (~150–300 µg/L free Hb); corresponds to ~5 RBC/HPF.
   - **Advantages:** Rapid (<2 min), inexpensive, high sensitivity (~91–100%).
   - **Disadvantages:** Cannot distinguish haemoglobin from myoglobin; false positives from oxidising agents.
2. **Urine microscopy (sediment analysis):**
   - **Principle:** Centrifuged urine sediment examined under microscope (400x HPF). RBCs appear as pale biconcave discs (~7 µm).
   - **Detection:** RBC count per HPF; morphology assessment (isomorphic vs dysmorphic).
   - **LOD:** ~1 RBC/HPF (practical).
   - **Advantages:** Differentiates isomorphic vs dysmorphic; identifies casts.
   - **Disadvantages:** Operator-dependent, semi-quantitative, time-consuming.

3. **Phase-contrast microscopy:**
   - **Principle:** Enhanced visualisation of RBC morphology for dysmorphic assessment (acanthocytes).
   - **Detection:** >5% acanthocytes = glomerular origin (sensitivity ~52–72%, specificity ~98%).
   - **LOD:** As for standard microscopy.
4. **Automated flow cytometry (Sysmex UF-5000):**
   - **Principle:** Fluorescent nucleic acid dye + flow cytometry; RBCs distinguished from WBCs and bacteria by scatter/fluorescence profile.
   - **Detection:** RBC/µL count.
   - **LOD:** ~5 RBC/µL.
   - **Advantages:** Rapid, reproducible, automated.
   - **Disadvantages:** Does not assess morphology; cannot distinguish dysmorphic.

Optimal Urine Type for Measurement

- **Midstream clean-catch specimen** — standard for microscopy and dipstick.
- **First morning urine** — concentrated, but crenated/lysed RBCs may be harder to identify; no clear advantage.
- Fresh sample is critical: analyse within **1–2 hours** (RBCs lyse in hypotonic/alkaline urine).
- **Do not refrigerate** for extended periods if microscopy planned (crystal formation obscures sediment).
- Avoid menstrual contamination in women (note timing).

Actual Gold Standard

Manual urine microscopy of centrifuged sediment remains the reference for RBC detection and morphology (isomorphic vs dysmorphic). Phase-contrast microscopy at 400x is the gold standard for dysmorphic RBC assessment. The dipstick is the most widely used screening tool, with microscopy for confirmation.

Interferences in Measurement

Interference	Effect	Mechanism
Myoglobin	False positive (dipstick)	Pseudoperoxidase activity similar to Hb
Free haemoglobin (haemolysis)	False positive for intact RBCs	Lysed RBCs release Hb; dipstick positive but no cells on microscopy
Ascorbic acid (vitamin C)	False negative (dipstick)	Reduces the chromogen, inhibiting colour development
Oxidising agents (hypochlorite)	False positive (dipstick)	Directly oxidise TMB
Menstrual blood	False positive	Contamination, not urinary bleeding
Alkaline/dilute urine	RBC lysis	Hypotonic lysis → ghost cells missed on microscopy; Hb detected by dipstick
Vigorous exercise	Transient haematuria	Benign; resolves in 24–72 h
Bacterial peroxidases	False positive (dipstick, rare)	Some bacteria produce peroxidases

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis spectroscopy of haemoglobin (Soret band):** Oxyhaemoglobin absorbs at **415 nm** (Soret), **542 nm**, and **577 nm** (Q bands). Methaemoglobin at 405 nm and 630 nm. Used for free Hb quantification in lysed urine.
  - LOD: ~0.5 mg/L free Hb (~10 RBC/µL if fully lysed).

  - Sample prep: Centrifuge to remove intact cells; measure supernatant.
- **NIR:** Limited utility — Hb has weak NIR absorption (overtones at ~760 nm for deoxyHb). Not validated for urine.

Fluorescence Detection

- **Autofluorescence:** RBCs have **weak autofluorescence** from porphyrin ring of haemoglobin (Ex 405 nm / Em 635 nm), but heavily quenched by iron. Not practically useful.
- **Fluorescent nucleic acid dyes (automated analysers):** Since mature RBCs lack nuclei, they show **no nucleic acid fluorescence** — this is how Sysmex UF differentiates RBCs from WBCs and bacteria. RBCs are identified by forward scatter (size) + side scatter + low fluorescence.
- **Zinc protoporphyrin fluorescence:** In reticulocytes/immature RBCs, ZnPP fluoresces (Ex 425 nm / Em 594 nm). Not used for urinary RBC detection.

Raman Detection RBC Raman spectrum is dominated by haemoglobin:

Peak (cm^{-1})	Assignment
~750	Pyrrole breathing (porphyrin)
~1003	Phenylalanine
~1223	Amide III
~1375	Oxidation state marker (ν_4)
~1550–1580	Spin state marker (ν_2, ν_{19})
~1620–1640	Vinyl C=C stretch

- Resonance Raman with excitation at **532 nm** (Q-band resonance) strongly enhances porphyrin modes; single-cell Raman is feasible.
- LOD: Single cell level (confocal); ~100 RBC/µL in bulk urine.
- SERS not typically needed due to resonance enhancement.
- **Sample prep:** Centrifugation to concentrate cells; glass slide/optical window.

FTIR Detection Relevant bands for RBCs/haemoglobin:

Band (cm^{-1})	Assignment	Notes
~1650	Amide I (protein secondary structure)	α-helix dominant in Hb
~1540	Amide II	N–H bend + C–N stretch
~1080	PO_2^- (membrane phospholipids)	—
~2850–2960	C–H stretches (membrane lipids)	Fatty acid chains

- ATR-FTIR on dried urine sediment can detect RBCs at ~10^3/µL. Not routinely used.
- **Sample prep:** Centrifuge urine, dry pellet on ATR crystal.

Voltammetry Detection

- **Haemoglobin electrochemistry:** Direct electron transfer of Hb at modified electrodes (carbon nanotubes, graphene oxide). Heme Fe^{3+}/Fe^{2+} redox couple at **~-0.35 V vs Ag/AgCl** (pH 7).
- LOD: ~0.1–1 mg/L free Hb in buffer.

- Not validated for intact RBC counting; detects free haemoglobin.
- **Sample prep:** Lysis to release Hb; dilution in buffer.

Other Detection Technologies

1. **Automated image analysis (Iris iQ200, Sysmex UN-Series):** Digital microscopy with AI-based particle classification. LOD: ~5 RBC/µL. No manual operator needed.
2. **Smartphone microscopy:** Portable lens attachments + ML classification of urine sediment images. LOD: ~10–50 RBC/µL. Emerging point-of-care tool.
3. **Electrical impedance (Coulter principle):** Cell counting by impedance changes as cells pass through an aperture. Used in haematology analysers; adaptable for urine. LOD: ~5–10 cells/µL.

Summary Table: Detection Methods Comparison

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Dipstick (TMB/peroxidase)	~5–10 RBC/µL	Green-blue colour (visual)	None	Rapid, cheap, sensitive	Cannot distinguish Hb/myoglobin
Microscopy (sediment)	~1 RBC/HPF	400x light microscopy	Centrifugation	Morphology, casts	Operator-dependent
Phase-contrast microscopy	~1 RBC/HPF	Phase optics	Centrifugation	Dysmorphic assessment	Specialised microscope
Flow cytometry (Sysmex)	~5 RBC/µL	Ex 488 nm / scatter	None	Automated, fast	No morphology
UV-Vis (Soret, free Hb)	~0.5 mg/L Hb	415 nm (Soret)	Centrifuge supernatant	Quantitative Hb	Misses intact RBCs
Resonance Raman	Single cell	Ex 532 nm; 750, 1375 cm^{-1}	Concentrate cells	Single-cell ID	Specialised
FTIR	~10^3 RBC/µL	1650, 1540 cm^{-1}	Dry pellet	Multianalyte	Low sensitivity
Electrochemical (Hb)	~0.1 mg/L Hb	-0.35 V vs Ag/AgCl	Cell lysis	Sensitive for free Hb	Destroys cells
AI digital microscopy	~5 RBC/µL	Digital imaging	None/auto	Automated, morphology	Instrument cost

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